

Joyee(Zhuoyi) Jin

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Skills

Languages: C++, C, Java, Python, JavaScript

FrameWork/ Technology: R, Microsoft Azure, NumPy, PyTorch, Express.js, Next.js, Git, CSS

Education

University of Toronto Sept 2023 – May 2028
B.Sc. in Computer Science and Computational Cognitive Science, Minor in Mathematics

Research Experience

Machine Learning Research Intern Hong Kong
Huawei Technologies, 2012 Laboratories Jul 2026 – Jan 2027
Leibniz Research Center, Theory and Optimization Control Lab

- Developed machine learning and reinforcement learning methods for wireless network optimization using real-world connectivity data.
- Built end-to-end Python pipelines for data processing, model training, validation, baseline comparison, and experimental analysis.
- Reduced unnecessary network switching by 81.3% compared with the guard-only controller.

AI Developer Intern California, US
LeadDigital May 2026 – Jul 2026

- Developed an AI workflow that converted technical ship blueprints and engineering drawings into structured outputs.
- Built data preprocessing and evaluation pipelines and analyzed model errors to improve output reliability.

Research Assistant California, US
Stanford University, Human-Computer Interaction in Artificial Intelligence May 2025 – Sept 2025
Feb 2026 – May 2026

- Developed and deployed Next.js and Vercel research prototypes to evaluate LLM prompt-engineering strategies in human-AI collaboration.
- Analyzed user-study data in R and applied machine learning models to support adaptive interface development.

Research Mentor Toronto, Canada
University of Toronto Mathematics Mentorship Program Nov 2025 – April 2026

- Mentored two high school students through clinical AI research, including study design, experimentation, model evaluation, and academic writing.
- Supervised their clinical LLM project, which was accepted to AIXMHC 2026.

Student Researcher Toronto, Canada
University of Toronto, IDELA Lab Sept 2025 – April 2026

- Developed machine learning models using large-scale APLUS educational data and evaluated AI-driven interventions and student outcomes.
- Supported the design and implementation of data-driven educational research methods.

Publications

- **Jin, J. W.**, Chen, Q., Mao, S., Zhang, Q., Wang, Y., Qian, S., & Zhang, F. “Learning When to Switch: Wi-Fi-to-5G Traffic Steering for Reliable Elevator Scenarios.” *Submitted to IEEE Wireless Communications and Networking Conference (WCNC 2027)*, 2026. (underreview)
- Dong, Z., Wang, B., & **Jin, J. W.** “Route, Don’t Fix: Regime-Dependent Decoding Correction and a Trajectory-Gated Router for Reliable Clinical LLM Answer Selection. *AIxMHC 2026*.Accepted
- Tang, X., **Jin, J. W.**, Li, X., & Cohen, G. “High Performance, High Risk? A Meta-analytic Audit (2022–2025) of Social-Media Depression Models and Their Ethical Gaps.” *The 2025 International Conference of the Computational Social Science Society of the Americas*, Santa Fe, NM, 2025.
- Tang, X., **Jin, J. W.**, Ma, E., & Li, X. “AI in the Wild: A Meta-Analytic Evaluation of Depression Detection from Social Media Data.” *AAAI 2026 Artificial Intelligence for Social Impact Track*, 2025.
- **Jin, J. W.**, Tang, X., Li, X., & Cohen, G. “Social Context in the Machine: Predicting Depression from Multimodal Online Behavior.” *SPSP*, 2025.

Projects

Paper Reference System

github.com/Paper-Reference-System 

- Built an academic writing tool with article summarization and semantic search using Python, reducing literature review time.
- Implemented document-processing and retrieval functionality to help users organize, search, and reuse information from research papers.

Airline Planning System (Python)

github.com/airline-RS 

- Implemented a flight planner using real Google Flights data, enabling custom itineraries and optimized search with Pandas/NumPy.
- Built filtering and ranking logic to compare flight options across routes, schedules, and user-defined travel constraints.

Hearing Disability Assistant

github.com/HOI 

- Designed an Arduino-based wearable that detects critical sounds such as alarms and knocks and logs alerts to a companion web app.
- Integrated hardware sensing with a web-based notification workflow to provide users with visual awareness of important environmental sounds.